

The effect of complexity on the optimal rate of trait mutation

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Abstract

This project explores how complexity affects the optimal rates of mutation in an evolving ecosystem. To examine this, I created a 2D model of multicellular life in which natural selection acted on agents or "bodies" whose traits were mutated directly. Different versions of the model involved different levels of body complexity: The 1st level consisted of single cells with fixed abilities and evolving neural networks; the 2nd level added evolvable abilities; the 3rd added evolvable multicellular bodies; the 4th added evolvable communication between cells within the same body. A number of tests confirmed the essential properties of the model: Ecosystems at later time-points were pitted against those from earlier time-points, and their superiority confirmed that non-local adaptation was occurring for each level. Ecosystems from higher complexity levels were pitted against those from lower levels, and their superiority confirmed that added evolutionary flexibility increased fitness. Overall, the results of these tests indicated that the model is an appropriate substrate for examining how complexity affects the optimal rate of evolution. I then found, for each level, the mutation rate parameters that produced the highest fitness. One parameter, the mutation rate of the weight matrix, was common to all levels. The optimal mutation rate for this parameter tended to increase with increasing model complexity, possibly because the larger evolutionary space created by increased complexity favors a faster search.

Introduction

Biological Evolution

There is overwhelming evidence that species as they exist today are the products of evolution that occurred over billions of generations (1, 2). This process can be most simply described by natural selection and mutation. Natural selection is the process by which organisms that are "more fit" to survive and produce offspring do so, while those that are "less fit" do not. By this process, the overall population will shift over time towards "more fit" organisms. Mutation introduces the possibility for evolution to produce new forms that had not existed prior. Mutation involves changes between the genetic makeup of the parent and the offspring, which leads to different phenotypes (traits) between the parent and the offspring. These new forms are also subject to natural selection, meaning that mutants that are "more fit" will end up outcompeting "less fit" mutants. Mutation allows evolution to increase the fitness of organisms far beyond the fittest individuals in the starting population.

Genetic Algorithms

Interestingly, these same principles of natural selection and mutation can be used in program optimization (3,4). There are many established methods for systematic optimization such as gradient ascent and linear optimization. Genetic algorithms are a comparatively recent method of optimization that has rapidly gained popularity.

The basic underlying principles that govern all genetic algorithms are generalizations of the underlying principles also governing biological evolution.

Genetic algorithms must contain "agents" or possible solutions that can be compared using heuristic functions and that can be modified via random changes to their components. In genetic algorithms, a heuristic function (5) is any way of determining which solution is "more fit" in terms of the desired results: a fitness function. For example, heuristic functions can be used to determine how well a particular set of parameter values match a given set of data, or how quickly a generated procedure arrives at a solution. Natural Selection can be viewed as an implicit heuristic function; the universe's fitness function preserves the genetic story of some agents, while discarding that of others.

Genetic algorithms must also contain some elements of random change or mutation; this allows for subsequent "generations" of agents to differ from the parent generations, and along with pruning using the heuristic function, mutation can lead to better agents/solutions (6). As a simple example, consider a genetic algorithm that has 100 agents; the heuristic function can be used to select the top 10 agents, and these 10 can each have 10 children copies made. Each child copy can then be given a small random change that makes each child unique, and pairs of children can have parts of their genomes swapped through recombination. The children become the 100 agents in the next generation (6). Ideally, the second generation will have better average heuristic function values than the first generation. This process can then be repeated until a desired level of solution is reached.

Genetic algorithms are extremely general in their potential applications. Any problem that consists of some starting variables and that has a heuristic function

can be explored with genetic algorithms. It may appear at first that genetic algorithms are a great universal method for optimization that will apply for almost any problem. However, there are several limitations of genetic algorithms that can lead to sub-optimal solutions, such as time and space constraints, as well as arriving at local maxima that do not reflect the optimal solution. Studying evolution allows us to improve the effectiveness of genetic algorithms and increase their accuracy and applicability.

The evolution of a system depends on several key factors. Perhaps the most important of these is modularity. In biology, modularity is the mapping of genes to traits so that the effects of altering a gene are largely restricted to a single function (3, 7). In this case, a function is any integrated system within an organism that operates relatively independently from other systems. Modularity allows functions to be optimized independently. The opposite of modularity is pleiotropy, in which a gene may affect multiple disparate functions. There are many examples of pleiotropy in nature, nonetheless, it is thought to be the exception (8). Modularity offsets the "cost of complexity" - the dramatic decrease in the rate of adaptation with the increase in organism complexity (9,10). The presented model assumes complete modularity, and the model components are mutated independently. This simplifying assumption is similar to the direct encoding approach used in studies of neuroevolution(11)

A second key design factor of evolvable systems is the mutation rate, and this factor is the focus of the thesis. The term "rate" signifies degree of change over time and so it combines both frequency and magnitude of change. If the mutation rate is

too slow for a given problem, then the rate of addition of positive changes may be too slow and an acceptable solution may never be developed. On the other hand, if the mutation rate is too fast, then any positive changes may be removed by mutation in subsequent generations, essentially undoing progress. This tradeoff is one essential consideration when choosing a mutation rate for a given genetic algorithm.

Mutation rate also affects one of the most problematic issues in genetic algorithms: the possibility of getting stuck at local maxima. If the solution space of the heuristic function has local maxima such that small mutations will not incrementally improve the solution, then the algorithm may get caught in these local maxima, rather than the true optimum. One way to address the problem of local maxima is to initialize the algorithm with a large starting population, so that many evolutionary paths can be explored simultaneously. The downside of this approach is that it may require considerable time and space. Another approach is to use a high mutation rate; it may be possible through larger mutations to leave local maxima and start climbing towards a better solution (12). However once again if the mutation rate is too high, it may be impossible to properly climb to any maximum at all because progress is undone by mutation, so once again a tradeoff exists.

A common genetic algorithm rule of thumb that has proven useful for many NP-hard problems is the $1/L$ guideline. The guideline promotes an inverse relationship between the size of a problem and the mutation rate per bit (13). This inverse relationship has also been observed in genomic evolution, as the mutation rate of smaller organisms (c-elegans, for example),

is larger per base pair than the rate of mutation for organisms with larger genomes (homo-sapiens) (14). While genome size does not necessarily correlate with complexity of the organism coded for, this is generally the case (15).

This background motivated the hypothesis tested in this paper: *The optimal mutation rate for a system is inversely related to the system's complexity.* This hypothesis was tested with a bio-inspired model of multicellular evolution, but the results may generalize to genetic algorithms as well. The remainder of the introduction is a brief overview of the bio-inspired model and the experiments that were performed to test the characteristics of the model and to test the hypothesis.

Model Summary

The model is an "Agent Based Model" meaning that it is composed of simulated agents, hereafter referred to as "Bodies". Bodies in the model exist in a 2D world and are able to move, detect and eat one another. Several versions of the model were tested, with each version corresponding to a different level of complexity. In levels 1 and 2, bodies are composed of a single cell. In levels 3 and 4, Bodies are given the evolutionary capacity to become multicellular. Each cell has inputs, such as the distance and angle to the closest cell that they see, etc. Cells also have outputs, such as where they intend to move, where they intend to look next, etc. Each cell's behavior is controlled by its own neural network. The neural network is essentially a mapping between inputs from the world, and the output decisions by the cell. At the 4th level of complexity, cells in the same Body are also

allowed to communicate, essentially by connecting an output of one cell to the input of another cell, allowing one-way passing of information.

The model runs in time units called ticks. Each tick involves several steps. These steps are described in detail below, but in brief: The first step involves the writing and processing of inputs and the execution of the resulting outputs. Outputs include actions like applying force to the cell's body, changing the cell's viewing direction, or taking energy from another body. Second, the body's energy level is either increased or decreased depending on its output actions. Third, bodies with 0 energy are removed and replaced with mutated copies of randomly chosen living bodies. After these three steps, the next tick cycle begins. When a 0-energy body is replaced by a mutated copy of a living body, the mutations that occur from parents to mutated children differ depending on the levels of complexity being studied. These mutations are outlined below, and detailed in the model description:

Table 1.

Mutations at each levels of complexity

Level 1	Weight Matrix	No Mutation	No Mutation	No Mutation
Level 2	Weight Matrix	Cell Abilities	No Mutation	No Mutation
Level 3	Weight Matrix	Cell Abilities	Body Formation	No Mutation
Level 4	Weight Matrix	Cell Abilities	Body Formation	Communication

Experiment Summary

Four experiments were run. The first two experiments employed the "Deathmatch Heuristic function" to test the evolutionary capacity of the different

levels of the model. This heuristic, modeled after (16), combines ecosystems from different evolutionary backgrounds and measures their ability to compete against each other. In the first experiment, the Non-Local Adaptation Test, "older" ecosystems (ecosystems that had run for more ticks) competed with "younger" ecosystems. If older ecosystems consistently outcompete younger ecosystems, this indicates that the model is capable of evolving. In the second experiment, the Inter-Level Comparison Test, the competition was between ecosystems of the same age but from different levels of model complexity. If ecosystems with potentially more complex bodies are able to outcompete ecosystems with less complex bodies, this indicates that the additional complexity confers additional evolutionary capacity.

The purpose of the first two experiments was to validate the appropriateness of the model for the third experiment: the Random Points Algorithm, which was used to test the hypothesis. This experiment sought, for each level, the optimal mutation rates for the parameters relevant to that level. (Note: The first two experiments were actually run after the third experiment using the parameters determined by the third experiment. For purposes of exposition, these validation experiments are described first.)

The two most complex levels, Levels 3 and 4, had multicellular bodies, but only Level 4 had the potential for cell-cell communication. Of course, having the potential for cell-cell communication does not guarantee that such communication evolved. To confirm that information was exchanged between cells in Level 4 bodies, a fourth experiment was conducted, the Cell-Cell Communication Mutual Information Test.

The methods and results of these experiments are explained in detail below. The next section describes the characteristics of the bio-inspired model that was used to conduct all four experiments.

Methods

Model Description

Model Motivation

The primary objective behind the design of the model was to create an abstract simulation of cooperative multicellular evolution driven by natural selection. The model begins with a completely randomized population of agents. Over time the progeny of these agents develop behaviors and body configurations that promote their survival. The model parameters were tuned with the intention of maximizing the variety of successful body sizes, shapes, and behaviors that develop.

Model Structures

As with any agent-based model, this program is composed of several data structure components that interact and change state to advance the system. These data structures will be referred to extensively in the rest of the model description. The following section provides an object reference.

Agent/Cell

Cells are the smallest single units of artificial life in the model. Each has 3 abilities, a 10x1 vector of neurons, and a 10x10 neuron weight matrix

Ability

Cells have three abilities, selected with replacement from: Move, Eat, and See. The Abilities are controlled by the cell's neural network. In levels 2, 3, and 4, abilities may change across generations through mutation.

Table 2.

Ability Descriptions

Note: there is a default metabolic cost of 20 for every cell, before any other costs are added

Ability Names	Metabolic Cost	Function	Inputs	Outputs
Move	5	Imparts a directional force to a point behind the center of mass of the cell, allowing the body to displace and rotate.	1) Dot product between current movement vector and ability force vector direction, input may help the cell monitor how the ability is contributing to the overall movement direction of the body.	1) Direction of force vector relative to cell heading 2) Magnitude of force vector
Eat	100	Creates an "eating straw" line extending from the center of the cell's body. If this eating straw overlaps with a cell from another body, 40000 energy will be moved from the other body to the eating body for	1) Energy amount to be consumed this tick (taste/touch, indicates whether the cell is successfully eating another cell)	1) Direction of "eating straw" relative to cell heading

		every tick that the overlap is maintained.		
See	15	Creates a "visual circle" some distance away from the cell, and will report information about the closest of any foreign bodies that it detects in the visual circle area.	1) Distance to closest foreign body (helps judge relative position of detected body) 2) Angle relative to cell heading of closest foreign body (helps judge relative angle of foreign body) 3) Energy contained in closest foreign body	1) Direction of visual circle drawing relative to cell heading 2) Distance visual circle will be drawn from body
Comm Input	0	Receives information from another cell in the same body every tick	1) Value written by Communication Output ability of another cell in body	None
Comm Output	0	Sends information to another cell in the same body every tick	None	Value read every tick and written to the Communication input ability of another cell in the body

Neural Network

Each cell has a 10-neuron network that controls its actions. Neurons have values written to them by ability inputs. The neuron values are then multiplied by the cell's 10x10 weight matrix, W , using the following equation: $[t+1] = S([t] \times W)$, where $[t+1]$ are the output neuron values, $[t]$ are the input neuron values, S is a parameterized sigmoid function (which keeps values between 0 and 1), and x is the dot product. After this step, neuron values are read to control cell outputs. The weight matrix is mutated between generations at all levels, with matrix values capped within the range -1 to 1.

Body

Bodies are either unicellular or multicellular. Multicellular bodies are bonded together and operate as one organism. Body sizes vary from 1 to 50 cells, but they rarely exceed a size of 20, given the chosen model parameters. Bodies are able to displace and rotate, but the constituent cells of the body have fixed relative positions. Bodies have a single shared energy value, when this value drops below 0, all cells in the body die.

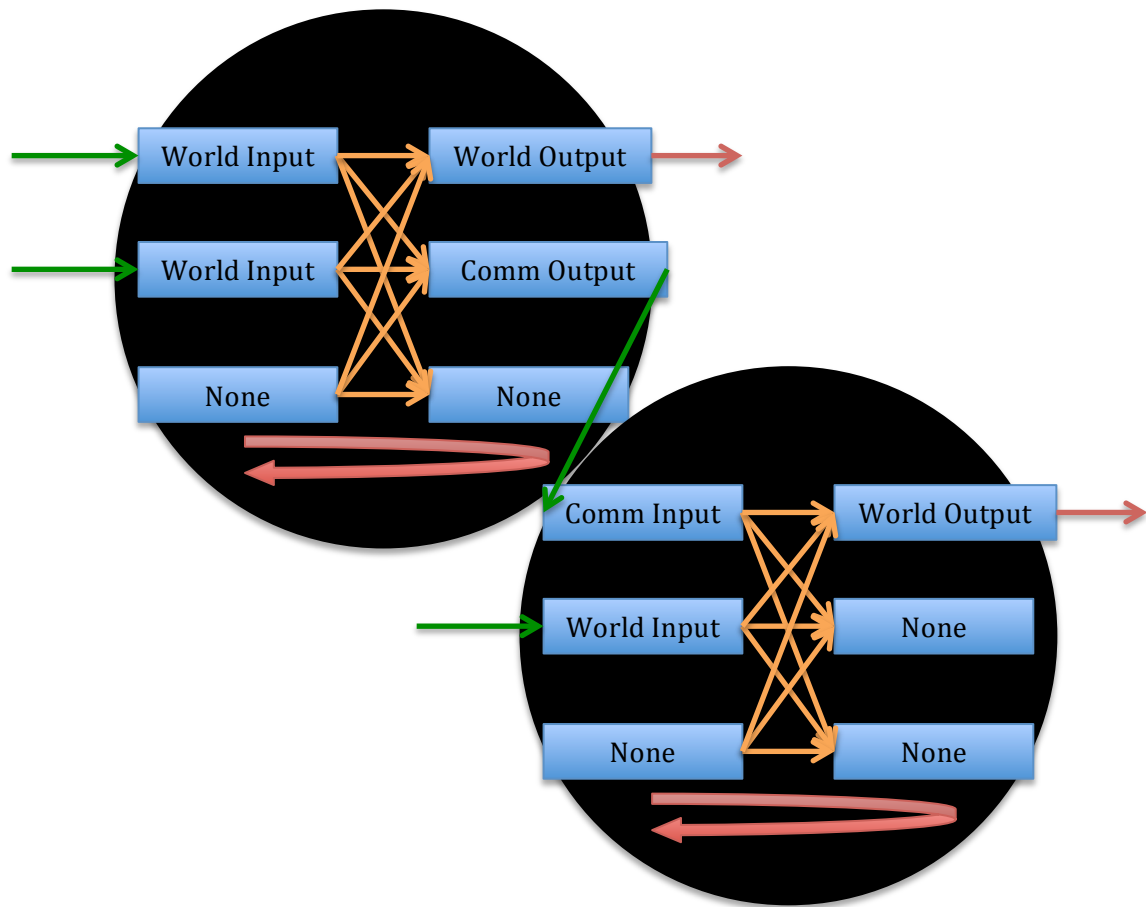
Cell-Cell Communication

Cell-Cell communication is introduced at the 4th level and pairs the Comm Input and Comm Output abilities between cells. When a 2-cell body is formed, 4 communication neurons per cell are randomly paired with the other cell as either inputs or outputs, producing 4 communication channels total (provided there are enough open neurons). When a new cell is added to the body, for each communication channel it will automatically either connect to any open communication neurons on the other cells in the body that can be used, or wait to connect with additional cells in the future. 4th level communication mutations can also alter the wiring scheme.

Tick Cycle

An atomic unit of time in the model is called a tick. During each tick, the following 6 events occur in order. The next step is not begun until all cells have finished the previous step, allowing all cells to operate with equal precedence.

- 1) Cell inputs from abilities are written to the neuron vector.
- 2) The input neuron vector is multiplied by the weight matrix of each cell to create the output neuron vector
- 3) The output neuron vector is read to determine cell actions for the next tick.
- 4) The physical simulation operates for one time step, cells carry out chosen actions
- 5) Bodies with energy < 0 are killed and removed (Selection System) and replaced with mutated copies of living bodies (Mutation System)



An example body diagram: a simplified diagram of a single body composed of 2 cells (black circles) 3 neurons per cell (rows of blue rectangles), and 1 communication line (diagonal green arrow). For simplicity, abilities are not explicitly represented. The left blue column in each cell shows the interpretation of the neurons during the input phase. The right blue column in each cell shows the interpretation of the neurons during the output phase. Green arrows indicate values that are written during the input phase, orange arrows indicate values that are read/written during the think phase, and red arrows indicate values that are read/written during the output phase, the red looping arrow indicates that previous output values become input values in the subsequent tick until they are overwritten by new inputs. Note that some neurons have no inputs. These may be utilized as memory, as they will maintain their output value from previous ticks.

Mutation System

The population initially starts with approx. 5,000 cells. The Mutation or "Reproduction" system replaces dead cells every tick. This system keeps the number of cells in the model approximately constant by replacing dead bodies with mutated children of living bodies. To replace a dead body, an exact copy is made of a living body and then random mutations are added as described in the mutation rate constants section. The child body is placed at a random position within the model.

Mutation Rate Constants

The mutation rate constants affect how different children bodies are from their parents. Each constant denotes the probability that a particular alteration will occur, or, in the case of the weight matrix, the magnitude of one such alteration. The model does not contain an explicit genome to but rather the weight matrix values, ability setups, and body configurations dictate body behavior and serve as an evolutionary record. This is known as a direct encoding, which bypasses the steps of creating and encoding a genotype to phenotype mapping. (17)

Level 1: Weight Matrix Mutation Strength

Every time a cell is reproduced, each entry of the weight matrix of the progeny cell is mutated in the following fashion:

- 1) A normally distributed random value with standard deviation equal to $e^{(\text{weight matrix mutation rate})}$ is added to the current weight value.
- 2) The weight matrix value is capped to be in the range (-1,1).

Level 2: Ability Mutation Probability

Every time a Cell is reproduced, a random real number in the range (0,1) is chosen for each ability the progeny cell possesses. If the random number is above $e^{(\text{Ability Mutation Probability})}$, then the ability will be swapped with a randomly chosen ability from the 3 possibilities (may result in no change).

Level 3: Probability Add or Remove Body Cell

Every time a Body is duplicated, a random real number in the range (0,1) is chosen. If it is above $e^{(\text{Probability Add or Remove Body Cell})}$, then another random number in the range (0,1) is chosen. If this number is below 0.5, a randomly selected cell on the periphery of the progeny body is removed. If the second number is above 0.5, a randomly chosen cell in the progeny body is cloned and the copy cell is permanently connected to the parent cell. The clone cell can evolve independently of the parent cell, allowing cells in the same body to differentiate over generations.

Level 4: Probability of Cell-Cell Communication Change

Every time a Body is duplicated, a random real number in the range (0,1) is chosen. If it is above $e^{(\text{Probability of Cell-Cell Communication Change})}$, then two pairs of cell-cell communication connections are randomly chosen, and these communication lines are cut and randomly rewired. This can lead to a changing of input/output directions, and can allow pairs of cells to communicate where no prior communication existed.

Natural Selection System

The process of "Natural Selection" is created by the inclusion of "energy" into the model. A cell has a fixed metabolic cost based on its abilities. The sum of all metabolic costs for all cells in a body is the per-tick metabolic cost of the body. There is also a per-cell cost for movement. These conditions cause all agents to lose some amount of energy every tick. Bodies can also "eat" each other, which transfers energy between bodies. Bodies die and are removed from the model when their energy drops below 0.

Energy Conservation

The interplay between the Natural Selection and Mutation systems keeps the total population nearly constant. However, to run the model indefinitely, the total energy of the model must also be kept constant.

A global count of used energy called the "SUN" was implemented to keep track of used energy. When a cell is randomly chosen to reproduce, $1/3$ of the energy of the parent is transferred directly to the child. The child energy is then doubled, with energy from the SUN. This causes the parent and child to have $2/3$ the energy that the parent had originally, which distributes more energy to children of parent bodies that were already doing well; expediting natural selection. The contribution to the child from the SUN was found through trial and error to assist the evolution of prey behavior, as it gives energy to lineages that would otherwise not receive any through hunting. To keep the energy in the model near constant,

residual energy in the SUN is distributed among child cells. In the A-Life model Tierra, there is a similar constant energy, in the form of allocated CPU time; this allows Tierra simulations to run indefinitely as well. (18)

Physics Simulation

To allow for body movement and collision in a semi-realistic and consistent manner, a simple 2D ball physics system was created. The goal of the collision system is to separate overlapping cells. This is done by applying a force proportional to the distance squared overlap between overlapping cells, so that they will be pushed apart with suitable force. The force is applied to the center of mass of the bodies of these cells as a combination of displacement and torque, which will cause bodies to bounce apart and rotate. The same system is used for the move ability, cells apply a combination of displacement and torque to the center of mass of their bodies when they use the move ability, which can cause their bodies to displace and rotate. It is by balancing these forces that bodies are able to move in a controlled manner.

Methods

Deathmatch Heuristic function

The Deathmatch Heuristic function was used in several experiments to compare the fitness of ecosystems with different evolutionary histories. To compare two ecosystems, the heuristic function randomly selects 2,500 cells-worth

of bodies from each ecosystem. The bodies are then randomly distributed into one world, with the default starting energy. For the duration of the comparison, mutation is turned off, meaning any progeny are direct clones of their parents. The model is run for 2,000 ticks. After every tick, the number of cells belonging to each ecosystem is tallied.

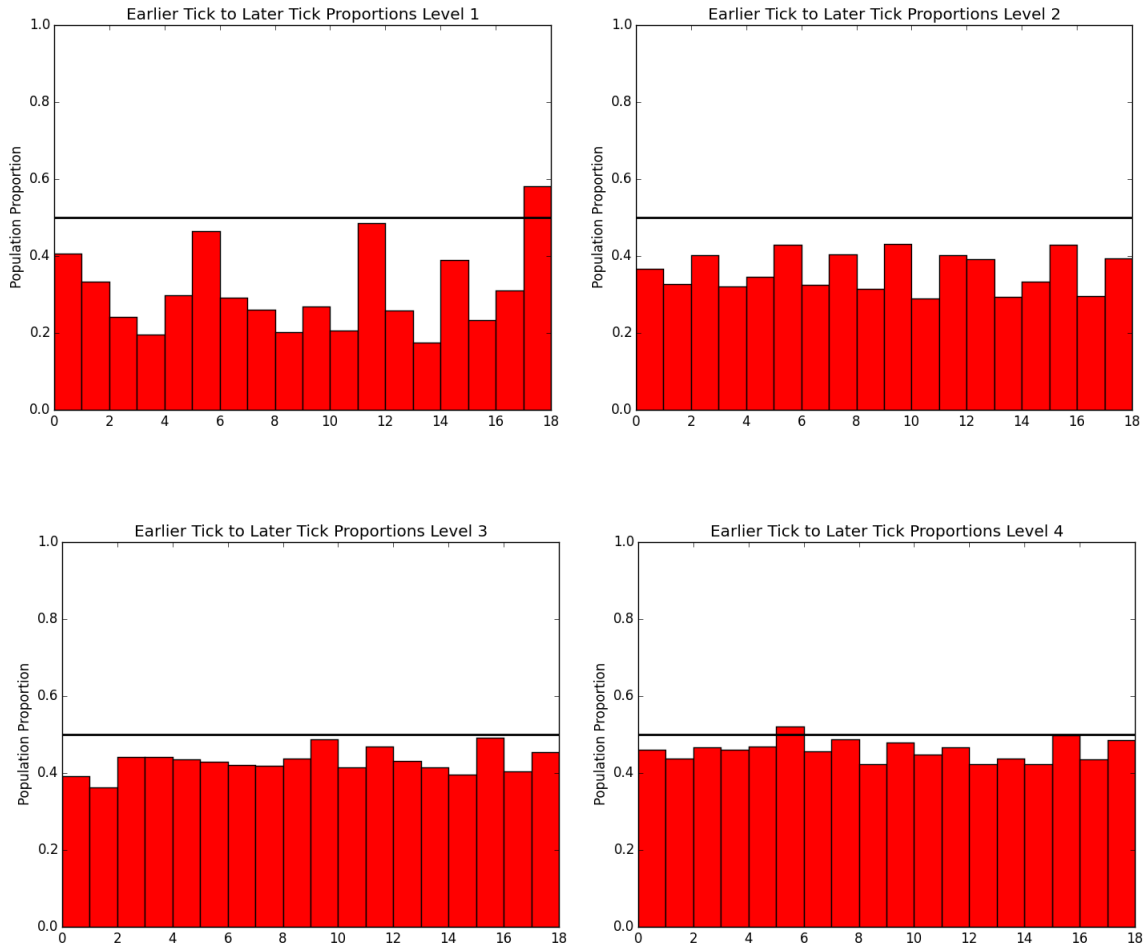
It should be noted that the Deathmatch Heuristic function does not provide an absolute measure of fitness. Each match involves a single model run with randomized starting positions of agents. Moreover, the matches provide only *relative* comparisons. The results cannot be used to rank ecosystems in order from better to worse. Instead, the comparison only gives a score between A and B, saying nothing about how either will do against a third ecosystem C.

Experiment 1: NON-LOCAL ADAPTATION TEST

The purpose of this experiment was to determine whether the model was capable of generalized adaptation. Testing local adaptation would be to sample a single model run at various timepoints, then pitting these against each other using the Deathmatch heuristic function to see how the timepoints compare. If later timepoints consistently beat earlier ones, then local adaptation is supported. The Non-Local Adaptation Test involves a similar comparison of timepoints, but does this comparison between ecosystems from different model runs. If later timepoints consistently beat earlier ones in the Non-Local case, then generalized adaptation has occurred, and older ecosystems are more highly evolved than younger ecosystems, regardless of their origin. Non-Local Adaptation has been observed in many

competitive evolving systems, such as models of iterated prisoner's dilemma playing agents (19,20,21), as well as models evolved to competitively paint a toroidal surface (22), and a grid-based predator prey model (23). Like the presented model, these models have agents evolving in a system with no direct measure of success beyond comparison to other evolving agents. This factor is important in discussing the non-local adaptation of a system, as evolving models without direct competition between agents (which are then measured purely via a heuristic function), feature only Non-Local Adaptation, as the agents in such models evolve in a vacuum and do not have to face local competition.

For each level of complexity, the model was run three times using the optimal parameters determined by the Rate Optimizing Genetic Algorithm. The resulting ecosystems were captured at ticks 2000, 4000, and 6000. Deathmatches were held between ecosystems derived from different runs, with a 2000 to 4000 tick difference between the compared ecosystems. Each panel shows the outcomes of 18 Deathmatches between an older population and a younger population from different ecosystems within the same level.



Level 1 (single cells with fixed abilities) Level 2(single cells with evolvable abilities) Level 3(single/multicellular bodies with evolvable abilities) Level 4(single/multicellular bodies with evolvable abilities and communication): Earlier (model runs frozen at 2000 and 4000 ticks) ecosystems are compared with later (model runs frozen at 4000 and 6000 ticks) ecosystems. Only ecosystems from different model runs were compared. Each red bar represents the earlier ecosystem population proportion from one ecosystem deathmatch comparison. If the earlier tick population proportion crosses 0.5, then agents from the earlier ecosystem outcompeted agents from the later ecosystem.

Results

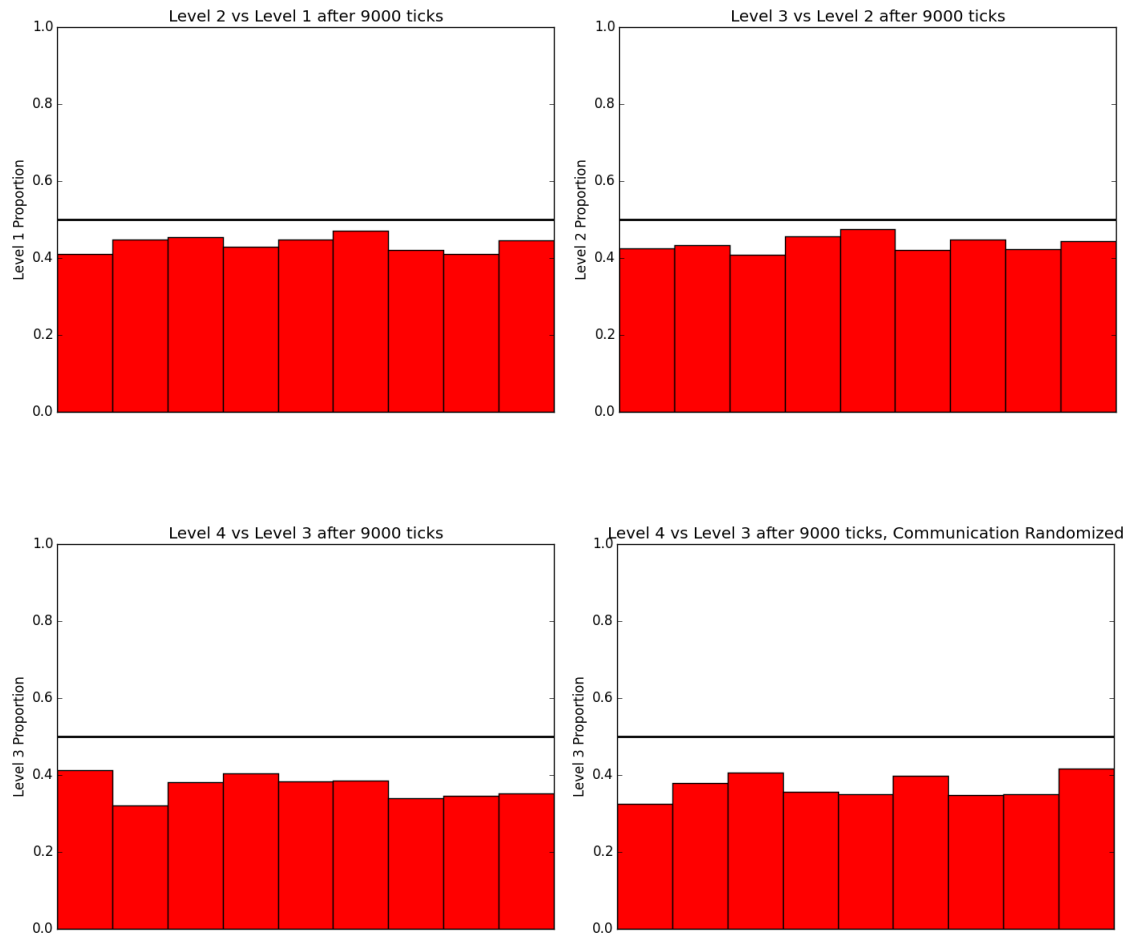
Interpretation of Non-Local Adaptation Results

The purpose of the Non-Local Adaptation Test was to demonstrate that evolution was occurring in the model. The four panels of Figure 1 support this claim. The bars represent the proportion of the total population belonging to the younger ecosystem. When this proportion is below 0.5, it indicates that the older ecosystems outcompeted the younger ecosystems. This occurs in all but 2 out of 72 cases, indicating that evolution is occurring at each level of the model (signed rank Wilcoxon test , $p = 9.46e-13$).

EXPERIMENT 2: INTER-LEVEL COMPARISON TEST

Experiment Description:

The Inter-Level Comparison Test was used to explore whether added levels of complexity are beneficial to ecosystem evolution, as well as whether relative benefits increase with evolutionary time.. This test is similar to the Test for Non-Local Adaptation in its logic and methodology: At each level, the model was run nine times and terminated at either 3000 ticks, 6000 ticks, or 9000 ticks. The resulting ecosystems were compared to those evolved for the same amount of time but at the next higher level of complexity. The scores from the Deathmatch Heuristic function were used as an index of "fitness", and again, it was assumed that winners were more highly evolved than losers. To assess the effect of cell-cell communication, (the property that distinguished levels 3 and 4) the Inter-Level Comparison Test was also run using levels 3 and 4, but in this case, communication input values sent between cells were replaced with uniformly distributed random numbers in level 4, referred to as randomized communication.



Level 1 (single cells with fixed abilities) Level 2(single cells with evolvable abilities) Level 3(single/multicellular bodies with evolvable abilities) Level 4(single/multicellular bodies with evolvable abilities and communication): ecosystems from models run for 9000 ticks were frozen at each level. Ecosystems from subsequent levels were compared for 2000 ticks. Lower level ecosystem population proportions over the 2000 ticks are shown as red bars. If the Lower level ecosystem population proportion crosses 0.5, then the lower level ecosystem outcompeted the higher level ecosystem. An additional experiment was later run in the same fashion comparing Level 3 ecosystems and Level 4 ecosystems with communication randomized.

Results

Interpretation of Inter-Level Comparison Test Results

In the model, levels are additive, with each higher level providing cells with more flexibility and features. Thus it was expected that evolved agents from later levels would outcompete those from earlier levels. This was observed in all comparisons between levels.

Of particular note is the finding that level 4 agents were able to consistently outcompete level 3 agents. It is tempting to take this as evidence of cell-cell communication, but an experiment described later suggests that information was not passed between cells in this version of the model. To better understand how Level 4 agents outcompeted Level 3 agents in this version of the model, an additional experiment was run comparing Level 3 to Level 4, but with communication input values from cells being substituted with uniform random real numbers:

Additional Inter-Level Comparison Experiment

In light of the findings from the Inter-Level Comparison Test and the Mutual Information Assessment of Cell-Cell Communication, an additional set of experiments were run similar to the Inter-Level Comparison Test but with randomized communication input values, (inserting a uniformly distributed random number between 0 and 1 instead of the communication output value from the communicating cell) to test whether communication output values were more advantageous than randomized communications. Surprisingly, random inputs appeared to be as useful to the cells as intact communication channels from other cells. The reason for this was not directly observed. It may be that bodies use these

channels (working or randomized) for perturbation to allow for more variation in their behavior.

The previous two experiments demonstrated that every level of the model is capable of evolving, and that higher levels of complexity can lead to higher levels of evolution. These findings provide necessary support for the next experiment, which is the primary focus of this thesis.

EXPERIMENT 3: RANDOM MUTATION SEARCH & WEIGHT MUTATION SEARCH

Experiment Description

The Random Mutation Search was used to determine the optimal evolution parameters at each level for different tick durations. The goal of the algorithm was to look for trends in these parameters across evolutionary time and levels of complexity. The search involved randomly sampling the space of mutation parameter values and comparing the resulting models using the Deathmatch Heuristic function as described above. The average of the best parameter values was used to approximate optimal mutation rates for different levels of model complexity.

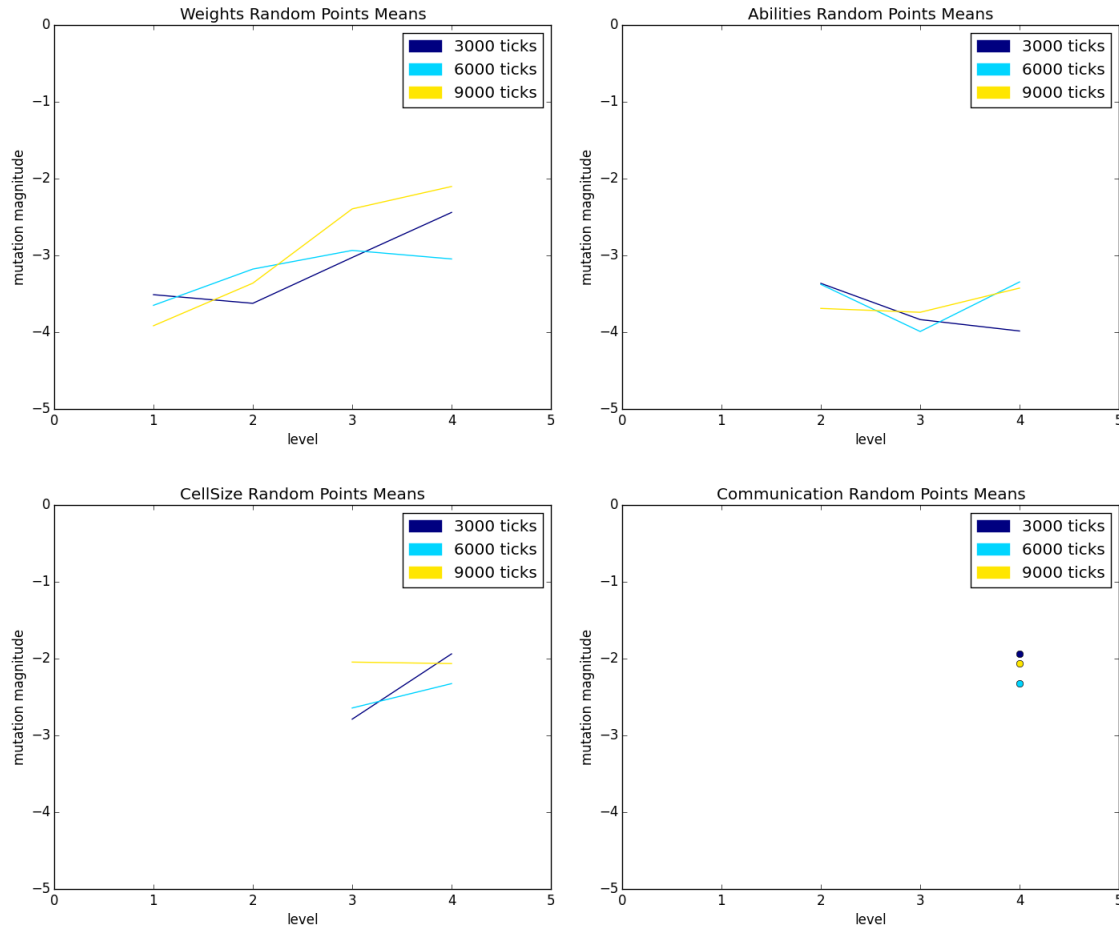
The details of the experiment were as follows: The four levels of complexity (Table 1), were each tested at 3000, 6000, and 9000 ticks. Each test involved 20 trials. Each trial involved a competition between four models with different evolutionary parameters randomly chosen on an exponential scale from $\exp(-5)$ to $\exp(0)$. A trial consisted of 6 runs, one for each possible pairing of the four parameters sets. The parameter set that yielded the highest scores for each of the 20 trials was averaged. This average was used as an estimate of the optimal parameter

values for that level and evolutionary time point. In all, the search involved 1,080 Deathmatches between 720 ecosystems. (6 deathmatches between 4 ecosystems *20 trials* 3 evolutionary time points * 3 levels of complexity).

Table 3.

Random Mutation Parameters Test Results

Tick Duration	Level	Dimension	Mean Value	Standard Dev
3000	1	0	-3.91596	0.79551
		1	-3.36156	1.02133
	2	0	-3.69049	0.86167
		1	-2.39440	1.49154
		2	-3.74064	0.84721
	3	0	-2.04546	1.46926
		1	-2.10266	1.48576
		2	-3.42558	1.01378
		3	-2.06469	1.25121
		4	-2.19307	1.28706
6000	1	0	-3.64981	0.79862
		1	-3.17807	1.20300
	2	0	-3.37732	0.93035
		1	-2.93497	1.51769
		2	-3.99102	0.73563
	3	0	-2.64312	1.33564
		1	-3.04623	1.39983
		2	-3.34563	1.52605
		3	-2.32569	1.49502
		4	-2.74385	1.58726
9000	1	0	-3.51140	0.66326
		1	-3.62380	1.14510
	2	0	-3.36384	0.88348
		1	-3.02724	1.39107
		2	-3.83565	0.77679
	3	0	-2.78850	1.39861
		1	-2.44069	1.47075
		2	-3.98429	0.95667
		3	-1.93862	1.33335
		4	-3.07390	1.45669



The Random Points Algorithm involves a random search through the mutation magnitudes. The value in each case is computed by $\exp(\text{mutation magnitude})$. Weight matrix mutation magnitudes set the standard deviation of Gaussian distributed random weight matrix value changes during reproduction. Ability mutation magnitudes set the probability of randomly mutating cell abilities during reproduction. Cellsize mutation magnitudes set the probability of randomly adding or removing a cell to/from a body during reproduction. Communication mutation magnitudes set the probability of separating two lines of communication between any two cells in a body, and then randomly rewiring their arrangement. Values represent the average of 20 winning values, with each winning value coming from a four-way competition. This method was repeated with comparisons of models run for 3000, 6000, and 9000 ticks.

Results & Discussion

Random Points Algorithm Results

The purpose of the Random Points Algorithm was to test the hypothesis that the optimal rates of evolution would slow down with increased evolutionary complexity. Given a range of length 5 and that points are chosen with uniform distribution, a totally random sample would have a standard deviation of around 1.44. Some of these results have standard deviations around this value, which could mean that they were less important to the results than the other parameters at that level to the point of irrelevance, or that there are multiple sinks for optimal values to occupy that are distant from each other. There is some consistency across all numbers of ticks in the positioning of the optimal points however, which lends some credibility to the resulting rate values. These results gave a starting set of parameter values that are used in the subsequent tests. Because the weight matrix mutation rates exist at all levels, these are the values that were used in the discussion.

WEIGHT MUTATION SEARCH

Experiment Description

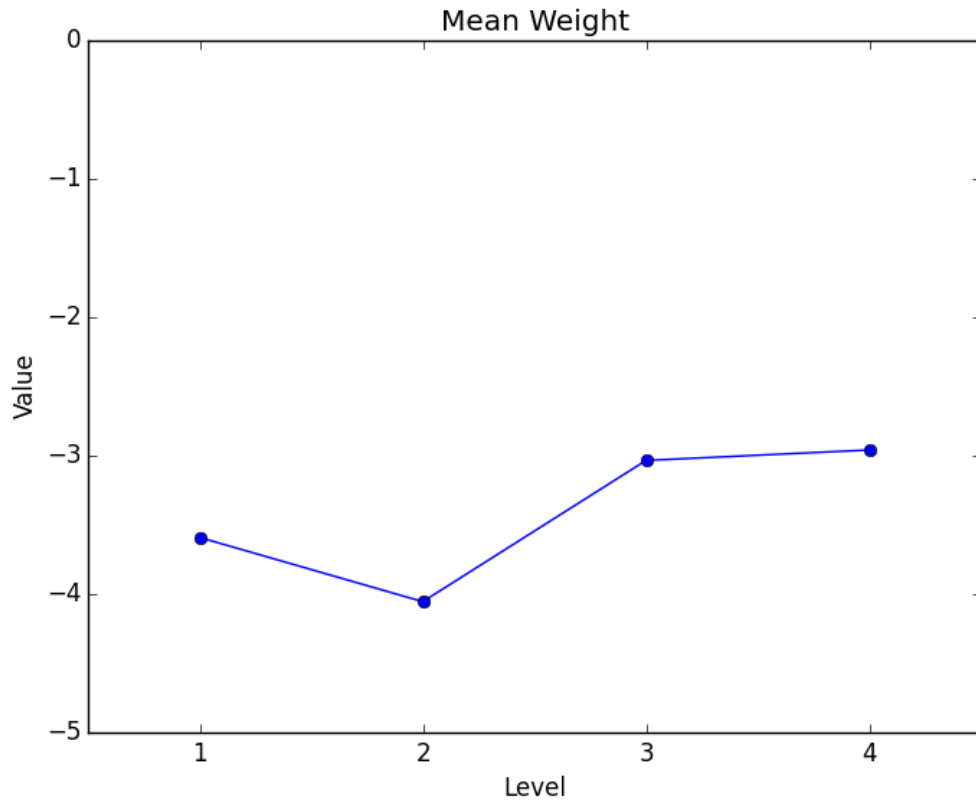
Because of the difficulty of simultaneously searching multiple parameter dimensions, a second search was conducted on the one parameter that was common to all four levels: the weight matrix mutation rate. This Weight Mutation Search was equivalent to the Random Mutation Search, except that only the Weight Matrix mutation rate changed. The other mutation parameters will be held constant at the optimal values previously estimated in the Random Mutation Search. The evolutionary time period was also held constant at 9000 ticks.

Here only one value was changed in order to assess how the weight matrix mutation values in isolation shift between levels.

Table 4.

Weight Mutation Search Results

Tick Duration	Level	Dimension	Mean Value	Standard Dev
9000	1	0	-3.59053	0.72538
	2		-4.05471	0.76619
	3		-3.03434	1.52584
	4		-2.95868	1.47447



Weight matrix mutation magnitudes set the standard deviation of Gaussian distributed random weight matrix value changes during reproduction, from $\exp(-5)$ to 1. Values represent the average of 20 winning values, with each winning value coming from a four way competition. Competing models were run for 9000 ticks. Other mutation magnitudes for levels 2-4 were set using the 9000 tick mean mutation magnitudes from table 3.

Results & Discussion

ANOVA results:

An ANOVA was run $F(3,76)=3.58$ with a resulting P value of 0.017647. all mean comparisons were insignificant except levels 2 vs 3 and 2 vs 4 meaning that the largest transition was between levels 2 and 3, with the addition of multicellularity.

Interpretation of Random Weights Results

Based on previous findings from biological and computational evolving systems, it was hypothesized that with each new level of complexity, the optimal mutation rate would slow down. This trend is not observed however, in fact the mutation rate of the weights tends to increase over generations. This puzzling trend seems to indicate that complexity and mutation rate are not so simply correlated. It may be that it depends on the nature of the added complexity, with some kinds of complexity being more permissive of rapid mutation than others. However one may speculate that the added complexity of clonal multi-cellularity may in fact increase mutation rates because differentiation becomes a driving factor that did not exist in the single cell case (the difference between levels 2 and 3).

It is noteworthy that there is very little difference between levels 3 and 4. The Inter-Level Comparison Test found level 4 outperforms level 3, but also happened when communication was randomized. So the final experiment will examine whether bodies have intercellular communication at level 4.

EXPERIMENT 5: MUTUAL INFORMATION TEST

Experiment Description

Mutual Information assessment of Cell-Cell Communication

As just noted, there were two multicellular versions of the model, level 3 and level 4. In level 3, the cells within a body could not communicate with each other, whereas in level 4, the cells could be linked by channels between their neural networks. To determine whether level 4 cells communicated information over these channels, I measured the mutual information between two pairs of information sources. One measurement pair consisted of each cell's sensory inputs (world-in) and its outputs to a different cell within the same body (comm-out). If cells evolved the capacity to relay information from their inputs to other cells in the body, this would increase the mutual information between world-in and comm-out. The other measurement pair was each cell's inputs from other cells (comm-in) and the cell's outputs to its environment (world-out). If cells evolved the capacity to use information from other cells to influence their activities, then this would increase the mutual information between comm-in and world-out.

To explain the mutual information measure, it is first necessary to define Shannon entropy, a measure of the randomness of an information source. Let the information source be a discrete random variable X . for all possible values x that X can take let $p_x = \text{Probability}[X=x]$. Then let the Shannon Entropy of X be defined as,

$$H(X) = - \sum_x \log_2(p_x) p_x$$

Mutual Information is a measure of the dependence between two information sources. Let X and Y be two discrete information sources. For all possible values x that X can take, and all possible values y that Y can take, Mutual Information is defined using the joint probability distribution of the random variables, $p_{x,y} = \text{Probability}[X=x, Y=y]$ as well as their individual probability distributions, $p_x = \text{Probability}[X=x]$ $p_y = \text{Probability}[Y=y]$:

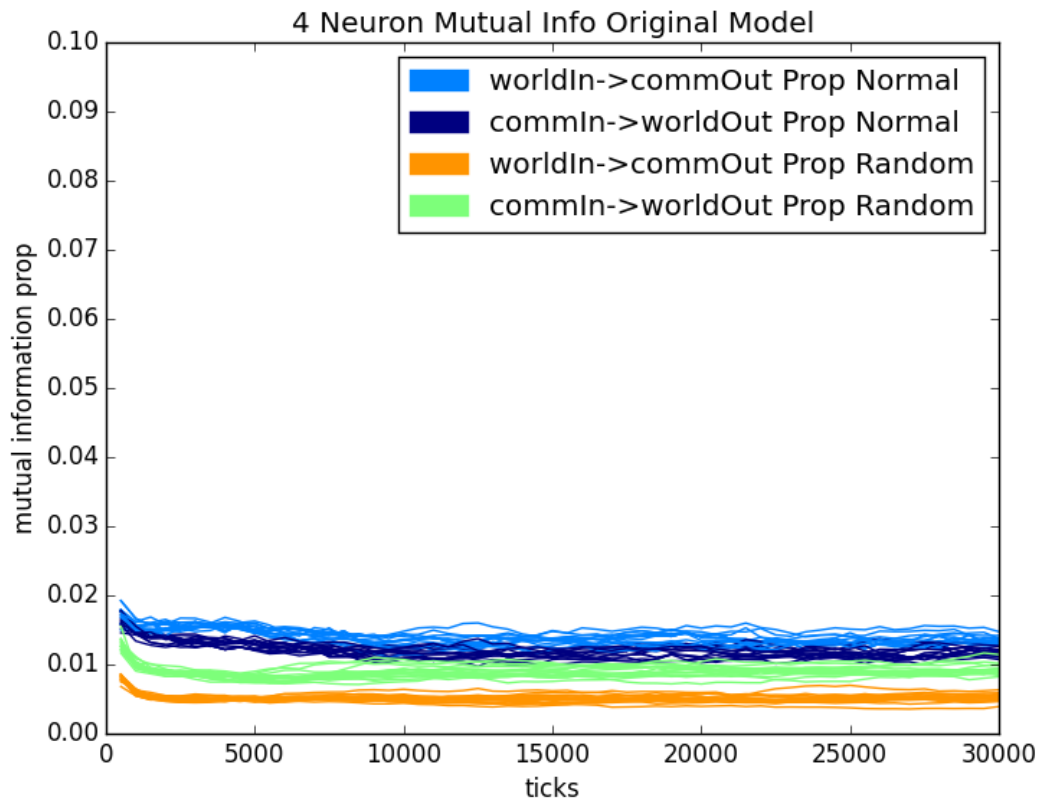
$$MI(X, Y) = \sum_{x,y} p_{x,y} \log_2 \frac{p_{x,y}}{p_x p_y}$$

To determine whether meaningful communication was developing at the 4th level, I examined the ratio of mutual information to total entropy in cell inputs and outputs. Mutual information was computed by binning results in 10 bins with 0.1 increments from 0 to 1. In measuring mutual information between world inputs and communication outputs for example, each input value would increment a bin count for that probability and each communication output value during that tick. This means that if a cell has multiple communication outputs and/or multiple world inputs, that these values will overlap in calculating the probability distributions for that cell. This was done for time and space constraints in collecting data, and was not expected to have major ramifications on the overall trends.

I examined how this proportion changed over time for cell inputs and communication outputs, as well as for communication inputs and cell outputs. If the bodies were evolving intercellular communication, then the proportion of mutual information to entropy should increase over evolutionary time compared to a control case where intercellular communication was randomized.

I ran the level model 16 times with communication working and communication randomized. Each run lasted 30,000 ticks, and every 500 ticks I sampled the entropy and mutual information of the intercellular communication. In particular, I measured the entropies of inputs from the world (world-in), outputs to the world (world-out), input communication (comm-in), and output communication (comm-out), and I measured the mutual information between world-in and comm-out, and between comm-in and world-out.

To compute the change in the mutual information proportion after these experiment runs, I ran an ANOVA to compare the first set of recordings (0-500 ticks) to the last set of recordings (29,500-30,000 ticks).



the mutual information proportion between cell inputs and outputs are calculated as the proportion of the entropy of two information sources that is “shared” or “explained” between them. This proportion was averaged for all cells every 500 ticks and plotted to observe the time evolution of 4 different mutual information sets. The higher this value, the more the two information sources are related. Randomized communication was also tested, in which cell output communication values were scrambled rather than based on values computed by integrating cell inputs.

Results and Discussion

ANOVA results:

An ANOVA was run $F(25,1)=448$ comparing the commIn->worldOut mutual information proportion for normal communication vs. random communication with a within subjects P value of <0.0001 . The mean value for normal communication lowered compared to the mean value for randomized communication.

Table 5.

commIn->worldOut ANOVA Results

Number Neurons	Start Mean Normal	End Mean Normal	Start Mean Random	End Mean Random	F Score	Cross P Value
4	0.017279	0.011688	0.00803	0.009198	(25,1)=448	<0.0001

An ANOVA was run $F(25,1)=424$ comparing the worldIn->commOut mutual information proportion for normal communication vs. random communication with a within subjects P value of <0.0001. The mean value for random communication lowered compared to the mean value for normal communication.

Table 6.

worldIn->commOut ANOVA Results

Number Neurons	Start Mean Normal	End Mean Normal	Start Mean Random	End Mean Random	F Score	Cross P Value
4	0.016279	0.013498	0.013111	0.00525	(25,1)=424	<0.0001

Additional Mutual Information Assessment of Cell-Cell Communication

Experiment Description

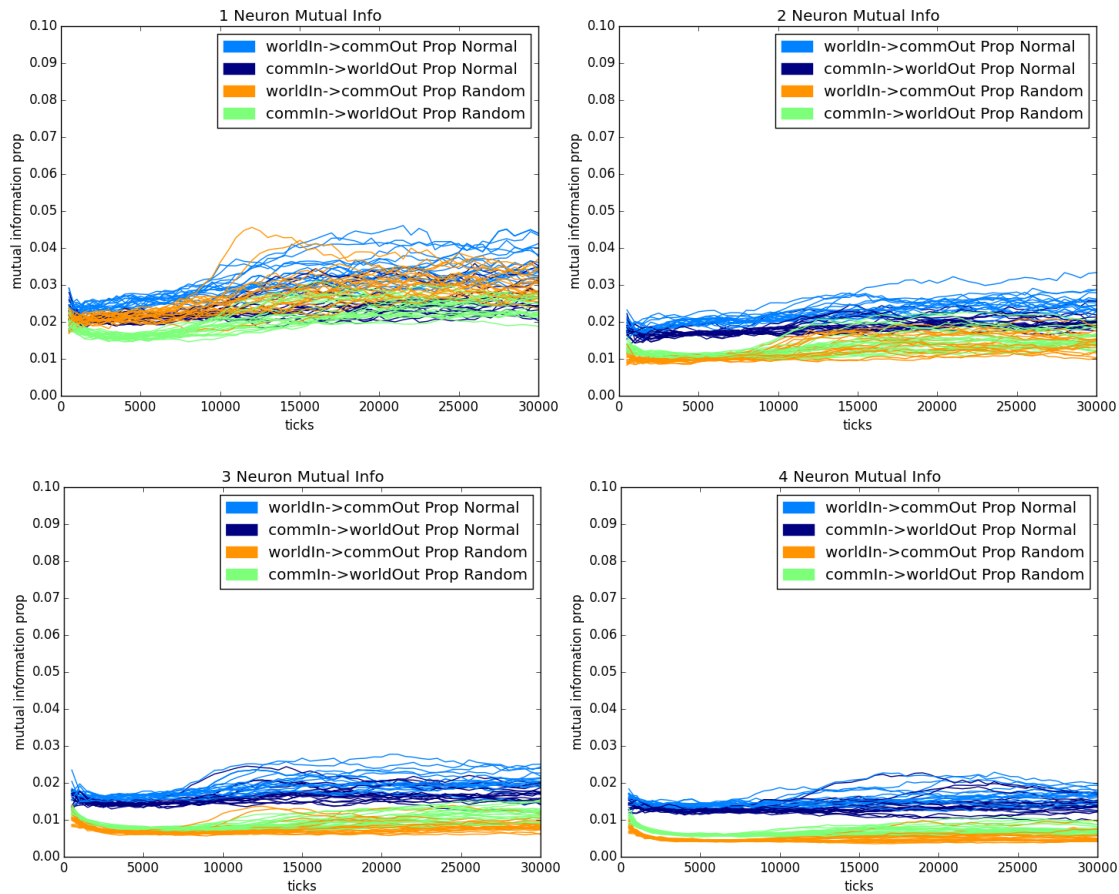
After gathering the results of the Additional Inter-Level Comparison Test, it was evident that randomized communication offered similar benefit as working communication in the initial version of the model. To increase the likelihood of generating working communication, the model was altered with changes made to some parameters, and the mutual information was measured with 1-4 communication neurons per cell. The following changes were made to the model:

- 1) The cost for having multiple different abilities was lowered from 60 energy per tick to 10 energy per tick to increase the likelihood of bodies integrating all 3 abilities

- 2) The size of the world was increased to promote usage of the See ability to find other agents and avoid predators
- 3) The costs of the 3 abilities were altered to ensure that they would be more evenly distributed.

An example run video of the resulting ecosystem was included as a demonstration of this model version. (24)

Plots of the mutual information proportion are shown with 1-4 communication neurons per cell.



A revised model version was measured for the mutual information proportion between cell inputs and outputs. Measurements were taken with bodies having from 1-4 communication neurons per cell to test if the mutual information proportion would be maximized with fewer communication channels.

Results & Discussion

Four ANOVAs were run comparing commIn->worldOut mutual information proportion for normal communication vs random communication with 1-4 Communication neurons per cell. P values were insignificant except for the 2 communication neurons per cell case with a P value of 0.00866. the means in this case were very close for normal communication, and increased for randomized communication.

Table 7.

commIn->worldOut ANOVA results

Number Neurons	Start Mean Normal	End Mean Normal	Start Mean Random	End Mean Random	F Score	Cross P Value
1	0.023453	0.026578	0.020616	0.024288	(33,1)=12.44	0.7208
2	0.019127	0.019843	0.010626	0.014919	(33,1)=191.5	0.000866
3	0.017245	0.016866	0.010636	0.011233	(33,1)=127.4	0.324813
4	0.015805	0.014	0.08026	0.07449	(33,1)=436.5	0.092895

Four ANOVAs were run comparing worldIn->commOut mutual information proportion for normal communication vs random communication with 1-4 Communication neurons per cell. P values were significant with P values <0.0001 except for the 1 communication neuron per cell case. The means remained similar or increased for working communication and decreased for randomized communication.

Table 8.

worldIn->commOut ANOVA results:

Number Neurons	Start Mean Normal	End Mean Normal	Start Mean Random	End Mean Random	F Score	Cross P Value
1	0.021933	0.034763	0.019906	0.029722	(33,1)=17.67	0.056967
2	0.01785	0.02585	0.013094	0.014223	(33,1)=212.6	<0.0001
3	0.015912	0.020812	0.012482	0.008738	(33,1)=255.5	<0.0001
4	0.013868	0.01633	0.0147	0.00546	(33,1)=432.5	<0.0001

Mutual Information Results Discussion

The purpose of the Mutual Information Test was to investigate whether cell-cell communication tends to develop over the duration of a model run. The results do not necessarily support this. The mutual information over entropy proportion graphs show that the mutual information between communication and

input/output remains relatively stable over the tick duration. The results are similar when communication is randomized, only with a decline during the first 5000 or so ticks. This decreased value is maintained for the rest of the model run. It can also be observed that when there is only 1 communication neuron per cell that the mutual information proportion is somewhat greater. The mutual information plots also show more spread when comparing the new model version to the original version, which may be because the new model version is more of a bottleneck of agent survival.

Based on these results, as well as the demonstrated utility of randomized communication in the level vs. level comparison, it may be that the bodies are utilizing the communication channels to add unstructured variation to their usual function, essentially injecting randomness into their behavior. While the utility of randomized inputs was not directly measured, one potential benefit is that this may allow agents that would otherwise move in circles to traverse the model in a varied fashion that covers more ground. The discrepancy in distribution between randomized communication (which is uniformly distributed) and working communication (which uses the cell state as input) may explain why the randomized communication proportion is less: it is a greater source of randomness. Additionally, in the case with a single communication neuron per cell, the neuron's significance may have been increased to compensate for the lack of other communication neurons promoting the randomization of behavior.

Another noteworthy result is that the mutual information proportion of worldIn->commOut separates between normal and randomized communication

(with normal increasing and randomized decreasing), while the opposite is observed for commIn->worldOut. The implications of these findings are unclear, but the results seem to indicate that cells are likely to evolve sending information from their inputs, but evolve against using that information to control their outputs (as compared to the randomized case).

General Discussion

The goal of this thesis was to assess how complexity affects optimal mutation rate. (The general trend was that optimal mutation rates sped up with increased complexity.) In the discussion we compare the model to other artificial life models, examine some important mutation concepts that were not mentioned, then address some of the results relative to the hypothesis. Finally some potential directions for future study are explored.

Comparison to Other Models

A similar model that investigates the emergence of multicellularity with neural network controlled individual agents that aggregate is Mosaic World (26). The model features a 2D grid with agents that can evolve to activate and deactivate abilities and thus differentiate. The model investigates possible reasons for the evolution of multicellularity, which the presented model takes as implicit assumptions, such as protection from predators and energy saving due to differentiation, and finds that these factors will tend to lead to multicellularity. The

presented model was built with assumptions based on these results by the incorporation of aggregate protection and faster aggregate movement as configurable constants (which were balanced in order to maximize the coexistence of many sizes of individuals).

A well-known model that explores artificial life while taking minimal influence from biology is called Tierra (18). This system creates digital organisms and uses CPU time as energy for these organisms. There is a spatial component as well, with the digital organisms competing with others at neighboring memory addresses. Tierra was not designed to mirror the behavior of organisms on an individual level, but does present emergent phenomena such as punctuated equilibria and host-parasite co-evolution. The model lacks long term evolvability, however (27). The presented model takes an approach that emphasizes a 2D simulated physics system to model some aspects of organism interaction, as it is imagined that an exploration of evolution may require modeling biological behavior as well as ecological trends.

A well-known model that shows many similarities is Evolving Virtual Creatures. (28) The model is similar in the sense that agents are constructed from body components with no preset body plan. The creatures both develop coordination through evolution of their morphology and their neural networks simultaneously. One key difference, however, is that the virtual creatures model has agents evolving without ecosystem competition. Instead agents are selected on the basis of their ability to complete specific tasks. Thus all evolution was geared towards a specific heuristic rather than the moving target of adaptive survival. The

model was also not geared towards studying evolution in and of itself, but was more of a proof of concept that such methods could produce viable simulated morphologies and behaviors.

Model Scale

While the model contains several aspects that are typical of many artificial life systems, such as the evolution of morphology through natural selection, what is unusual about the model is its scalability. Open problems in artificial life (25) states that one of the principle challenges of artificial life is to "Create a Formal Framework for Synthesizing Dynamical Hierarchies at All Scales". The scales referred to are ecosystems, organisms, organs, tissues, cells, organelles, molecules, and atoms. The presented model attempts to bridge this scale spectrum from the functional unit of the cell, to the overall ecosystem, and demonstrates attributes of each of these levels. Agents can exist in isolation as single cells, or as "tissues or organs" within multicellular organisms, and compete for survival in ecosystems that contain multiple species and developmental trends. The ecological shifts were not directly measured, but can be observed in a typical model run. The relevance of multicellularity on these ecosystems is exemplified in the comparisons between levels 2 and 3.

Long Term Evolvability Criteria

Evolvability can be defined as the capability of a system to generate adaptive phenotypic variation and to transmit it via an evolutionary process. (27) If a

heuristic function returns an accurate number, the evolvability of a system can be considered the rate of change of the heuristic function values measured after subsequent time points or generations of that system.

The long term evolvability of the model was not studied beyond the first 6000 ticks. While the model does show non-local adaptation up to that point, the time when this becomes inconsistent remains undetermined. The model is a decent candidate for long term evolvability according to the four criteria given in (29), which are the following:

1) Agents must meet a nontrivial minimal criteria to reproduce: This minimal criteria is having non-zero energy

2) Evolution of new individuals should create novel opportunities for satisfying the minimal criteria: New morphologies and neural networks provide this

3) Decisions about how and where individuals interact with the world should be made by the individuals themselves: this criterion is met because agents are able to see very far from themselves and change their movement direction based on this information

4) The potential size and complexity of the individuals' phenotypes should (in principle) be unbounded: This criterion is met because agents are able to form multicellular bodies. While the size of these bodies is bounded (currently set to 50), bodies don't normally go so far, and tend to stay within 20 or so cells with the current settings, while exploring a nearly infinite number of spatial configurations.

The meeting or approaching of all of these criteria by the model supports its potential for long-term evolvability. Maximizing long-term evolvability and effective cooperation between agents remain as significant artificial life challenges.

Unconsidered Evolution Concepts

An important assumption made by this paper is that mutation rates are fixed. In biological evolution as well as genetic algorithms this is not necessarily the case. In biological evolution DNA is checked with correcting proteins to fix errors. Depending on how often such proteins are employed (which is also genetically controlled) the natural mutation rate of DNA can be artificially inflated or mitigated, using mechanisms that evolve along with the rest of the DNA. (14) Changing mutation rates can also be useful for Genetic Algorithms. A convergence method known as simulated annealing is often used in genetic algorithms to help with the mutation rate issues presented above. This process can be applied to genetic algorithms by slowing down the rate of mutation over the duration of the solution search. This deals somewhat with the local maximum problem by at first giving the evolving agents flexibility to explore the solution space, and then ideally narrowing in on the best solutions.

Recombination and crossing over are also common phenomena in biological evolution and genetic algorithms. (6) These processes work by recombining partial solutions and/or genomes, rather than simply randomly mutating atomic components, and are very important to understanding the processes of biological evolution as well as for the construction of better genetic algorithms. Higher-level

recombination has been somewhat worked into the model with the adding and removing clone cells from bodies, but was not a central focus of the study.

Mutation Rate Results Relative to Expectations and Other Literature

Contrary to expectation, the mutation rates in the model have generally sped up as complexity has increased, despite the underlying assumption based on biology that larger genomes evolve more slowly. This unexpected finding forces a more thorough breakdown of added complexity. While every additional level does add some complexity in terms of increasing the number of evolutionary branching paths that the organisms can take in the model, it may be important to make a distinction between complexities that independently increase the evolutionary search space, and those that require coordination between these complexities. In the case where there are simply more evolutionary decisions to be made that do not require coordination, it may be that a faster rate of mutation is beneficial because it allows for a faster exploration of the larger evolution space. However in the case where coordination is necessary, it may be beneficial to slow down the mutation rate to allow for such coordination. The tradeoff between exploring possible configurations and maintaining a stable species is known as exploration vs exploitation (30) and is an established yet poorly understood aspect of evolutionary algorithms.

The biological analogue to this trend is the expectation that larger genomes would have faster mutation rates. This is exemplified by some plant species, as some genuses will produce large sequences of noncoding DNA and repeated sequences, and will be prone to rapid mutation of these regions (31,32). Analogously, It could

be that simply having a larger search space to explore in the multicellular case spurred the faster evolution rates observed (33), and that the increased complexity and redundancies may have made multicellular bodies more robust to deleterious mutations (34).

The $1/L$ mutation rate guideline (35) contradicts this theory, and suggests that a good rule of thumb for specifying the mutation rate for a genome/problem of size L is inversely proportional to that size (or $1/L$). The presented model avoids this issue of scaling mutation rate with problem size (or size of bodies) by mutating weights and abilities relative to cells rather than to bodies. Thus the effective mutation rate of a body is the mutation rate of an individual cell times the mutation rate of the body.

The increased search space may not be simply due to the increased number of cells per body, but may instead be due to an increase in the number of potentially successful configurations, as well as the necessity to exploit these configurations within a model run. Exploration is thought to be necessary in the presented model because as bodies evolve from single cells, the combination of forces that cause stable movement must be continually readjusted to allow bodies to navigate effectively. Cells must also adjust their reactions to inputs from detection of other cells to respond effectively. The most successful cellular responses change as bodies evolve, which may require faster mutation rates to reach them.

FUTURE DIRECTIONS

The concept behind the design of the model was to recreate a simplified model of multicellular evolution. While the model results generally support that the model is evolving in a somewhat characterizable fashion, it is still difficult to tease out specific aspects of the evolution of the model to study them in isolation. Subsequent experiments might be conducted on an even simpler version of the model, perhaps on only one of the 4 levels, and exploring evolutionary phenomena from that perspective. This would allow for more direct assumptions about the causality of the results observed, and could allow for a more focused approach depending on the question being asked.

Further exploring how mutation rates are affected by complexity could possibly yield additional insight into how phenomena like cloning cells into bodies affects mutation. Cloning to form bodies could be compared to attachment by cells from other lineages, which may have some impact on mutation rate.

The present study also raises many questions about how the cells are using communication to promote their survival. The mutual information results do not shed much light on the usefulness of communication, leaving room for more work. Applying a cost for communication and making the ability optional may offer some insight, as might comparing models with randomized communication to those with working communication directly through the Deathmatch Heuristic function. Another possibility could be to compare models with a high average mutual information proportion to those with a lower proportion value to see if the proportion indicates non-local evolutionary superiority.

A model that looks at the evolution of multicellularity from an individual organism perspective is Games of multicellularity (36). Their model uses a game theory approach, with cells within an “organism” operating as either cooperators or defectors in the hawk-dove and prisoner’s dilemma games. The presented model contains implicit cooperation, but it may in the future be interesting to remove this assumption to observe the ecological conditions that promote defectors and cooperators if cells are given the option.

The model was built with the assumption that new abilities may be added for additional study as well, and the interface for this is rather flexible. The only facets of the model that could not be altered by changing abilities are the 2D walled design of the model, the reproduction system, and the physics system. This means that the model could be used in the future to assess a variety of potential agent based evolutionary questions, possibly in the context of some specific agent communities, that may give a very different context than the general multicellular one explored here. Additional flexibility is gained with the structs.h file, which allows for virtually any numerical parameter that is baked into the model design to be modified, which greatly increases the number of potential tests that could be run.

It may also be interesting use the model to study development. In the present model, agents are born immediately in a complete multicellular form. This could be altered to instead allow for development to happen dynamically with agents cloning as buds from each other and either remaining attached or detaching depending on the instructions in their genome, which would also evolve. This would allow for interesting lifelike morphologies and growth strategies to develop. (37)

The model also has the capacity to have different terrain tiles added to change aspects of the model based on position. This would allow for the study of niche formation

Another interesting aspect to study is what factors most influence the growth of multicellular organisms. This could be facilitated by adjusting various model parameters and assessing how this affects the distribution of differently sized bodies that are produced.

It may also be possible to study different neural network topologies outside the simple fully connected case presented. The NEAT/HyperNEAT algorithms could be utilized to promote the dynamic generation of neural topologies (38)(24) as such a method could be an essential component to increasing the evolvability of agent behavior.

References

- (1) Darwin, Charles. "On the Origin of Species." (1859).
 - (2) Bonner, J. T. 1988. *The Evolution of Complexity by Means of Natural Selection*. Princeton Univ. Press, Princeton.
 - (3) Gunter P. Wagner, Lee Altenberg. 1996. Perspective: Complex Adaptations and the Evolution of Evolvability. *Evolution*. 50(3)
 - (4) Mitchell, M. "An introduction to genetic algorithms Cambridge, MA." (1998).
 - (5) Hart, Peter E., Nils J. Nilsson, and Bertram Raphael. "Correction to a formal basis for the heuristic determination of minimum cost paths." *ACM SIGART Bulletin* 37 (1972): 28-29.
 - (6) Whitley, Darrell. "A genetic algorithm tutorial." *Statistics and computing* 4.2 (1994): 65-85.
- APA
- (7) Wagner, Günter P., Mihaela Pavlicev, and James M. Cheverud. "The road to modularity." *Nature Reviews Genetics* 8.12 (2007): 921-931.
 - (8) Wagner, Günter P., and Jianzhi Zhang. "The pleiotropic structure of the genotype-phenotype map: the evolvability of complex organisms." *Nature Reviews Genetics* 12.3 (2011): 204-213.

- (9) Orr, H. Allen. "Adaptation and the cost of complexity." *Evolution* 54.1 (2000): 13-20.
- (10) Fisher, Ronald (1930). [*The Genetical Theory of Natural Selection*](#). Oxford, UK: Oxford University Press.
- (11) Yohannes Kassahun, Mark Edgington, Jan Hendrik Metzen, Gerald Sommer and Frank Kirchner. Common Genetic Encoding for Both Direct and Indirect Encodings of Networks. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO 2007), London, UK, 1029-1036, 2007.
- (12) Ewert, Winston. "Overabundant mutations help potentiate evolution: The effect of biologically realistic mutation rates on computer models of evolution." BIO-Complexity 2015 (2015).
- (13) Ochoa, Gabriela. "Setting The Mutation Rate: Scope And Limitations Of The 1/L Heuristic." GECCO. 2002.
- (14) Paul D. Sniegowski, Philip J Gerrish, et.al. "The evolution of mutation rates: separating causes from consequences." *Bioessays* 22.12 (2000): 1057-1066
- (15) Haygood, Ralph. "Mutation rate and the cost of complexity." *Molecular biology and evolution* 23.5 (2006): 957-963.
- (16) David Doty. Non-local Evolutionary Adaptation in GridPlants. *Bioinformatics and Computational Biology*: Iowa State Univ.
- (17) Kassahun, Yohannes, et al. "A common genetic encoding for both direct and indirect encodings of networks." Proceedings of the 9th annual conference on Genetic and evolutionary computation. ACM, 2007.
- (18) Ray, Thomas S. "Evolution, ecology and optimization of digital organisms." *Santa Fe* (1992).
- (19) Ashlock, Dan, and Brad Powers. "The effect of tag recognition on non-local adaptation." *Evolutionary Computation*, 2004. CEC2004. Congress on. Vol. 2. IEEE, 2004.
- (20) Ashlock, Dan, and John E. Mayfield. "Acquisition of general adaptive features by evolution." *Evolutionary Programming VII*. Springer Berlin Heidelberg, 1998.
- (21) Ashlock, Daniel, Eun-Youn Kim, and Wendy Ashlock. "Fingerprint analysis of the noisy prisoner's dilemma." *Evolutionary Computation*, 2007. CEC 2007. IEEE Congress on. IEEE, 2007.
- (22) Ashlock, Dan, Elizabeth Blankenship, and Jonathan Gandrud. "A note on general adaptation in populations of painting robots." *Evolutionary Computation*, 2003. CEC'03. The 2003 Congress on. Vol. 1. IEEE, 2003.
- (23) Ashlock, Daniel, and Adam Sherk. "Non-local adaptation of artificial predators and prey." *Evolutionary Computation*, 2005. The 2005 IEEE Congress on. Vol. 1. IEEE, 2005.
- (24) Pugh, Justin K., and Kenneth O. Stanley. "Evolving multimodal controllers with HyperNEAT." Proceedings of the 15th annual conference on Genetic and evolutionary computation. ACM, 2013.
- (25) Bedau, Mark A., et al. "Open problems in artificial life." *Artificial life* 6.4 (2000): 363-376
- (26) Schlessinger, Ehud, Peter J. Bentley, and R. Beau Lotto. "Investigating the emergence of multicellularity using a population of neural network agents." *Parallel Problem Solving from Nature-PPSN IX*. Springer Berlin Heidelberg, 2006. 711-720.

- (27) Hu, Ting, and Wolfgang Banzhaf. "Evolvability and speed of evolutionary algorithms in light of recent developments in biology." *Journal of Artificial Evolution and Applications* 2010 (2010): 1.
- (28) Sims, Karl. "Evolving virtual creatures." *Proceedings of the 21st annual conference on Computer graphics and interactive techniques*. ACM, 1994.
- (29) Soros, L. B., and Kenneth O. Stanley. "Identifying necessary conditions for open-ended evolution through the artificial life world of chromaria." *ALIFE 14: The Fourteenth Conference on the Synthesis and Simulation of Living Systems*. Vol. 14. 2014.
- (30) Eiben, Agoston E., and Cornelis A. Schippers. "On evolutionary exploration and exploitation." *Fundamenta Informaticae* 35.1-4 (1998): 35-50.
- (31) Sloan, Daniel B., et al. "Rapid evolution of enormous, multichromosomal genomes in flowering plant mitochondria with exceptionally high mutation rates." *PLoS Biol* 10.1 (2012): e1001241.
- (32) Genome Evolution: Unexpected relationship between mutation rate and genome complexity
- (33) Lehman, Joel, and Kenneth O. Stanley. "Evolvability is inevitable: increasing evolvability without the pressure to adapt." *PloS one* 8.4 (2013): e62186.
- (34) Lenski, Richard E., et al. "Genome complexity, robustness and genetic interactions in digital organisms." *Nature* 400.6745 (1999): 661-664.
- (35) Thesis Sample Run at Unlisted YouTube url:
<https://youtu.be/40ItYMegVP8>
- (36) Kaveh, Kamran, Carl Veller, and Martin A. Nowak. "Games of multicellularity." *Journal of theoretical biology* 403 (2016): 143-158.
- (37) Bongard, Josh. "Morphological change in machines accelerates the evolution of robust behavior." *Proceedings of the National Academy of Sciences* 108.4 (2011): 1234-1239.
- (38) Stanley, Kenneth O., and Risto Miikkulainen. "Evolving neural networks through augmenting topologies." *Evolutionary computation* 10.2 (2002): 99-127.